

SAT Math Mastery · Week 8B

Circles: Equations, Arcs, Tangents & Radians

Time 120 minutes: 90 min solving + 30 min self-check and error log

Topics circle equation · completing the square · center/radius from expanded form · circle position tests · arc length & sector area · central & inscribed angles · tangent $\perp$ radius · radians & degrees · unit circle

Name / Date _____

1 · SKILLS & IDEAS

- S1.** Read the center and radius from $(x - h)^2 + (y - k)^2 = r^2$, and write the equation from a center and radius, a diameter, or a graph.
- S2.** Complete the square to convert $x^2 + y^2 + ax + by + c = 0$ to standard form; use $r^2 > 0$ to find a constant.
- S3.** Translate or enlarge a circle and track the center and r^2 .
- S4.** Decide whether a point is on, inside or outside a circle; use tangency to an axis ($|\text{center coordinate}| = r$) or to another circle (distance between centers $= r_1 + r_2$ or $|r_1 - r_2|$).
- S5.** Compute arc length and sector area from a central angle in degrees or radians, and work backward to the radius or angle.
- S6.** Use the central-angle and inscribed-angle theorems (inscribed angle $= \frac{1}{2}$ central angle on the same arc; angle in a semicircle is 90° ; opposite angles of an inscribed quadrilateral are supplementary).
- S7.** Use “a tangent is perpendicular to the radius at the point of tangency”, including equal tangent segments from an outside point.
- S8.** Convert between radians and degrees; evaluate $\sin$, $\cos$, $\tan$ at standard angles on the unit circle and use quadrant signs.

2 · REQUIRED FACTS

- The graph of $(x - h)^2 + (y - k)^2 = r^2$ is a circle with center (h, k) and radius r (the right side is r^2 , not r).
- A point is inside the circle if its distance to the center is less than r , on it if equal, outside if greater.
- A circle is tangent to the x -axis when $|k| = r$, and to the y -axis when $|h| = r$.
- A full circle is $360^\circ = 2\pi$ radians, so $\pi \text{ rad} = 180^\circ$. An arc is the fraction $\frac{\theta}{360^\circ}$ (or $\frac{\theta}{2\pi}$) of the whole circle.
- An inscribed angle is half the central angle that intercepts the same arc. An angle inscribed in a semicircle is a right angle.
- A tangent line is perpendicular to the radius drawn to the point of tangency; two tangent segments from one outside point are equal.
- On the unit circle the point at angle θ is $(\cos \theta, \sin \theta)$; the signs follow the quadrant (All, Sine, Tangent, Cosine going counterclockwise).
- Common values: $\sin 30^\circ = \frac{1}{2}$, $\sin 45^\circ = \frac{\sqrt{2}}{2}$, $\sin 60^\circ = \frac{\sqrt{3}}{2}$; $\cos$ is the same list in reverse order.

3 · REQUIRED FORMULAS

$$(x - h)^2 + (y - k)^2 = r^2, \quad x^2 + y^2 + ax + by = c \Rightarrow \left(x + \frac{a}{2}\right)^2 + \left(y + \frac{b}{2}\right)^2 = c + \frac{a^2 + b^2}{4}$$

$$s = r\theta \text{ (}\theta \text{ in radians)} = \frac{\theta^\circ}{360} \cdot 2\pi r, \quad A_{\text{sector}} = \frac{1}{2}r^2\theta = \frac{\theta^\circ}{360} \cdot \pi r^2 = \frac{1}{2}rs, \quad C = 2\pi r, \quad A = \pi r^2$$

$$\text{rad} = \text{deg} \cdot \frac{\pi}{180}, \quad \text{deg} = \text{rad} \cdot \frac{180}{\pi}, \quad \text{inscribed} = \frac{1}{2}\text{central}, \quad d(P_1, P_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

4 · MINI THEORY SHEET

Equation of a circle. Group x -terms and y -terms, add $\left(\frac{\text{coefficient}}{2}\right)^2$ to *both sides* for each variable, then read h , k and r^2 . Signs flip: $(x+3)^2$ means $h = -3$. If a problem gives the radius, set the right-hand side equal to r^2 and solve for the constant.

Which formula for arcs and sectors?

Angle given in	Use
degrees	fraction $\frac{\theta}{360}$ of $2\pi r$ (arc) or of πr^2 (sector)
radians	$s = r\theta$ and $A = \frac{1}{2}r^2\theta$
arc and sector area known	$A = \frac{1}{2}rs$ gives r without knowing θ

Angles in a circle. Central angle = its arc (in degrees). Inscribed angle = $\frac{1}{2}$ its arc. Same arc $\Rightarrow$ equal inscribed angles. Draw radii to make isosceles triangles ($OA = OB = r$).

Tangents. Radius $\perp$ tangent gives a right triangle with legs r and the tangent length, and hypotenuse from the center to the outside point. Tangent lengths from one point are equal.

Radians. Convert with $\pi \leftrightarrow 180^\circ$. Co-terminal angles differ by 2π . A point at angle θ on a circle of radius r is $(r \cos \theta, r \sin \theta)$; moving clockwise subtracts the angle.

Speed shortcuts.

Situation	Shortcut
radius from $x^2 + y^2 + ax + by + c = 0$	$r^2 = \frac{a^2+b^2}{4} - c$
center from expanded form	$\left(-\frac{a}{2}, -\frac{b}{2}\right)$
circle tangent to an axis	the radius equals that center coordinate
diameter endpoints	center = midpoint, $r = \frac{1}{2} \cdot \text{distance}$
two sectors with equal area	$r_1^2\theta_1 = r_2^2\theta_2$
arc and radius given	$\theta = \frac{s}{r}$ rad, no degrees needed
tangent from an outside point	$\text{length}^2 = d^2 - r^2$
radius doubled/tripled	r^2 is multiplied by 4 or 9

5 · GUIDED EXAMPLES

Example 1 — Complete the square

The graph of $x^2 + y^2 + 6x - 8y - 11 = 0$ is a circle. Find its center and radius.

Group: $(x^2+6x)+(y^2-8y) = 11$. Add $\left(\frac{6}{2}\right)^2 = 9$ and $\left(\frac{8}{2}\right)^2 = 16$ to both sides: $(x+3)^2+(y-4)^2 = 11+9+16 = 36$. Center $(-3, 4)$, radius $\sqrt{36} = 6$.

Example 2 — Arc length and sector area

A circle has radius 9. A central angle measures 80° . Find the arc length and the sector area.

In radians, $80^\circ = 80 \cdot \frac{\pi}{180} = \frac{4\pi}{9}$. Arc: $s = r\theta = 9 \cdot \frac{4\pi}{9} = 4\pi$.

Sector: $A = \frac{1}{2}r^2\theta = \frac{1}{2}(81)\frac{4\pi}{9} = 18\pi$. (Check: $\frac{80}{360} \cdot \pi(81) = 18\pi$.)

Example 3 — Tangent, radius and inscribed angle

$\overline{PA}$ is tangent to a circle with center O and radius 7 at A , and $OP = 25$. Also, an angle inscribed in the circle measures 35° . Find PA and the central angle on the same arc.

$OA \perp PA$, so $PA = \sqrt{25^2 - 7^2} = \sqrt{576} = 24$ (the triple 7-24-25).

Central angle $= 2 \cdot 35^\circ = 70^\circ$.

6 · PRACTICE SET — 30 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1. EASY · NO CALCULATOR NEEDED

The equation of a circle in the xy -plane is $(x - 3)^2 + (y + 5)^2 = 49$. Which of the following gives the center and the radius of the circle?

(A) center $(-3, 5)$, radius 49

(B) center $(3, -5)$, radius 7

(C) center $(-3, 5)$, radius 7

(D) center $(3, -5)$, radius 49

2. EASY · NO CALCULATOR NEEDED

Which of the following is equal to $\frac{5\pi}{6}$ radians?

(A) 135°

(B) 300°

(C) 150°

(D) 120°

3. EASY · NO CALCULATOR NEEDED

A circle has a circumference of 20π . The area of the circle is $k\pi$. What is the value of k ?

4. EASY · NO CALCULATOR NEEDED

What is the value of $\sin \frac{5\pi}{6}$?

(A) $\frac{1}{2}$

(B) $-\frac{1}{2}$

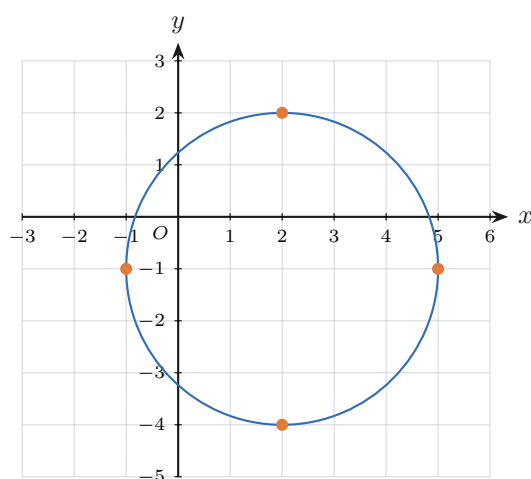
(C) $\frac{\sqrt{3}}{2}$

(D) $-\frac{\sqrt{3}}{2}$

5.

EASY · DESMOS

The graph of a circle in the xy -plane is shown. Which of the following is an equation of the circle?



(A) $(x + 2)^2 + (y - 1)^2 = 9$

(B) $(x - 2)^2 + (y + 1)^2 = 3$

(C) $(x - 2)^2 + (y - 1)^2 = 9$

(D) $(x - 2)^2 + (y + 1)^2 = 9$

6.

MEDIUM · NO CALCULATOR NEEDED

In the xy -plane, the graph of $x^2 + y^2 - 10x + 4y + 13 = 0$ is a circle. What is the radius of the circle?

7.

MEDIUM · NO CALCULATOR NEEDED

A circle has a radius of 12 centimeters. An arc of the circle has a central angle of 75° . What is the length, in centimeters, of the arc?

(A) 30π (B) 75π (C) 5π (D) 10π **8.**

MEDIUM · NO CALCULATOR NEEDED

Which of the following points lies on the circle with equation $(x + 1)^2 + (y - 4)^2 = 25$?

(A) $(1, -4)$ (B) $(2, 8)$ (C) $(-2, 7)$ (D) $(3, 0)$ **9.**

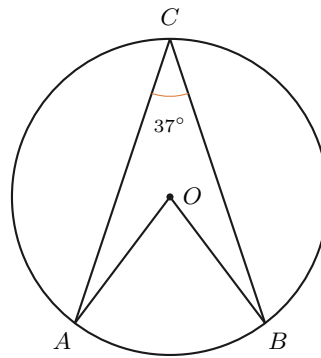
MEDIUM · NO CALCULATOR NEEDED

A circular sector has a radius of 6 and a central angle of $\frac{2\pi}{3}$ radians. The area of the sector is $k\pi$. What is the value of k ?

10.

MEDIUM · NO CALCULATOR NEEDED

In the figure, A , B and C lie on the circle with center O . What is the measure, in degrees, of $\angle AOB$?



(A) 37

(B) 143

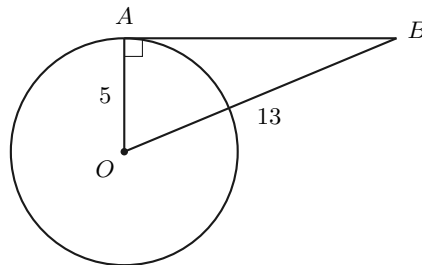
(C) 106

(D) 74

11.

MEDIUM · NO CALCULATOR NEEDED

In the figure, $\overline{AB}$ is tangent to the circle with center O at point A . What is the length of $\overline{AB}$?



(A) 12

(B) 8

(C) $\sqrt{194}$

(D) 18

12.

MEDIUM · NO CALCULATOR NEEDED

A circle has a radius of 24. An arc of the circle has length 10π . What is the measure, in degrees, of the central angle that intercepts this arc?

13.

MEDIUM · NO CALCULATOR NEEDED

The endpoints of a diameter of a circle in the xy -plane are $(-2, 3)$ and $(6, -5)$. Which of the following is an equation of the circle?

(A) $(x - 2)^2 + (y + 1)^2 = 16$

(B) $(x - 2)^2 + (y + 1)^2 = 32$

(C) $(x - 2)^2 + (y + 1)^2 = 128$

(D) $(x + 2)^2 + (y - 1)^2 = 64$

14.

HARD · NO CALCULATOR NEEDED

In the xy -plane, the graph of $x^2 + y^2 - 8x - 6y + k = 0$ is a circle that is tangent to the x -axis, where k is a constant. What is the value of k ?

15.

HARD · NO CALCULATOR NEEDED

In the xy -plane, two different circles are each tangent to both coordinate axes, lie in Quadrant IV, and pass through the point $(9, -2)$. What is the sum of the radii of the two circles?

16.

HARD · NO CALCULATOR NEEDED

The circle $(x + 2)^2 + (y - 3)^2 = 20$ is translated 4 units to the left and 5 units down. Then the radius of the translated circle is tripled, with the center unchanged. Which of the following is an equation of the resulting circle?

(A) $(x - 2)^2 + (y - 8)^2 = 180$

(B) $(x + 6)^2 + (y + 2)^2 = 540$

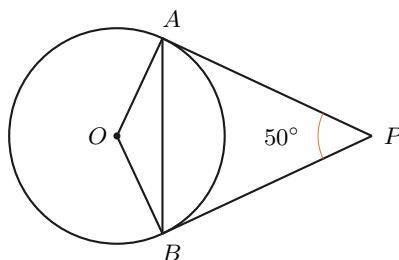
(C) $(x + 6)^2 + (y + 2)^2 = 180$

(D) $(x + 6)^2 + (y + 2)^2 = 60$

17.

HARD · NO CALCULATOR NEEDED

In the figure, $\overline{PA}$ and $\overline{PB}$ are tangent to the circle with center O at A and B , and $\angle APB = 50^\circ$. What is the measure, in degrees, of $\angle OAB$?



(A) 25

(B) 40

(C) 50

(D) 65

18.

HARD · NO CALCULATOR NEEDED

A circular sector has an area of 40 square centimeters and an arc length of 10 centimeters. What is the radius, in centimeters, of the circle?

19.

HARD · NO CALCULATOR NEEDED

For an angle θ with $\pi < \theta < \frac{3\pi}{2}$, $\sin \theta = -\frac{5}{13}$. What is the value of $\tan \theta$?

(A) $-\frac{5}{12}$

(B) $\frac{12}{5}$

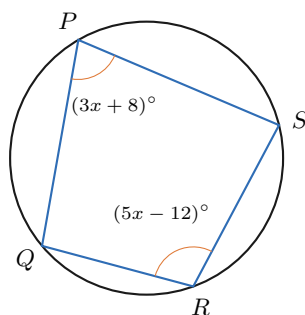
(C) $-\frac{12}{5}$

(D) $\frac{5}{12}$

20.

HARD · NO CALCULATOR NEEDED

Quadrilateral $PQRS$ is inscribed in a circle. The measures of angles P and R are $(3x + 8)^\circ$ and $(5x - 12)^\circ$, respectively. What is the measure, in degrees, of angle R ?



(A) 115

(B) 77

(C) 103

(D) 23

21.

HARD · CALCULATOR

A circular running track has a radius of 0.12 kilometers. A runner starts at one point on the track and runs along the track through a central angle of 200° . To the nearest meter, how far does the runner run?

22.

HARD · NO CALCULATOR NEEDED

A circular pizza with a radius of 9 inches is cut into slices, and one slice has a central angle of 40° . A second circular pizza has a radius of 12 inches. A slice of the second pizza has the same area as the 40° slice of the first pizza. What is the central angle, in degrees, of the slice of the second pizza?

(A) 71.1

(B) 22.5

(C) 30

(D) 53.3

23.

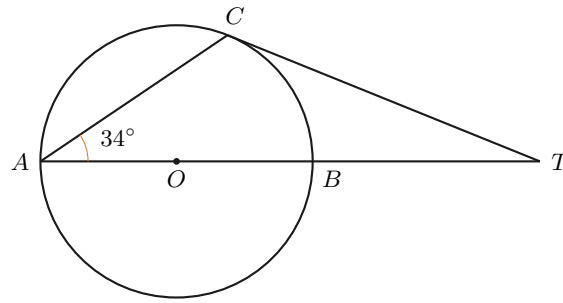
HARD · NO CALCULATOR NEEDED

In the xy -plane, the graph of $x^2 + y^2 - 6x + 2ky - 39 = 0$ is a circle whose center lies on the line $y = x + 1$, where k is a constant. What is the radius of the circle?

24.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the figure, $\overline{AB}$ is a diameter of the circle with center O , point C lies on the circle, and line CT is tangent to the circle at C with T on line AB . What is the measure, in degrees, of $\angle CTA$?



(A) 34

(B) 56

(C) 68

(D) 22

25.

SAT CHALLENGE · NO CALCULATOR NEEDED

A circular sector has a perimeter of 20 centimeters (two radii plus the arc) and an area of 24 square centimeters. The central angle of the sector is θ radians. What is the sum of all possible values of θ ?

26.

SAT CHALLENGE · NO CALCULATOR NEEDED

The line $3x + 4y = k$, where k is a positive constant, is tangent to the circle $x^2 + y^2 - 4x + 6y - 12 = 0$ in the xy -plane. What is the value of k ?

(A) 19

(B) 25

(C) 31

(D) -31

27.

SAT CHALLENGE · NO CALCULATOR NEEDED

Circle A is given by $x^2 + y^2 - 2x - 4y - 11 = 0$ and circle B is given by $x^2 + y^2 - 14x + 12y + c = 0$ in the xy -plane, where c is a constant. The two circles are externally tangent (they touch at exactly one point and neither lies inside the other). What is the value of c ?

28.

SAT CHALLENGE · NO CALCULATOR NEEDED

A point (a, b) lies on the unit circle at angle θ radians in standard position, where $0 < \theta < 2\pi$ and $ab > 0$. Which of the following must be true?

I. $\tan \theta > 0$ II. $\sin(\theta + \pi) \cos(\theta + \pi) > 0$ III. $\sin \theta > 0$

(A) I only

(B) I and II only

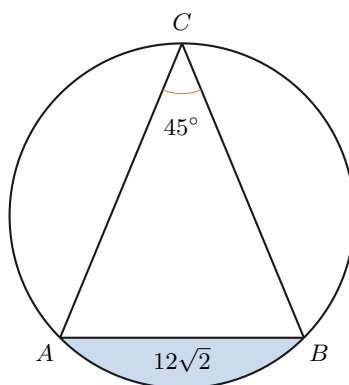
(C) II and III only

(D) I, II, and III

29.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the figure, A , B and C lie on a circle, $\angle ACB = 45^\circ$, and $AB = 12\sqrt{2}$. The area of the shaded region, bounded by chord $\overline{AB}$ and the arc AB that does not contain C , is $a\pi - b$, where a and b are positive integers. What is the value of $a + b$?



30.

SAT CHALLENGE · NO CALCULATOR NEEDED

A point starts at $(6, 0)$ and moves clockwise along the circle $x^2 + y^2 = 36$ a distance of 11π . What are the coordinates of the point where it stops?

(A) $(3\sqrt{3}, -3)$ (B) $(-3\sqrt{3}, 3)$ (C) $(3\sqrt{3}, 3)$ (D) $(3, 3\sqrt{3})$